Ludovic Bergeron

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Experience

Applied ML Research Intern April 2026 – August 2026
Christian Dior Couture (AI Factory) *Paris, France*

- Owned an open-ended R&D project end to end: framed the problem statement and research direction myself from loosely defined business needs, with no predefined objective.
- Surveyed the recommendation literature and implemented a Prod2Vec / Meta-Prod2Vec recommender (Skip-gram with Negative Sampling) as a first baseline.
- Engineered product metadata as side information (pairing exploratory visual analysis with product-domain expertise) and benchmarked configurations under a rigorous offline protocol (HR@K, NDCG@K, MRR, coverage@K) to isolate the attributes that most improved ranking quality.

Supervised Research September 2026 – January 2027
esiealab (ESIEA), Advisor: Dr. Mélanie Piot *Paris, France*

- Making deep-RL controllers for type 1 diabetes insulin delivery explainable through causal (anti-causal) inference: inferring which factor (meal, stress, hormonal cycle) drives each glucose deviation and feeding it back as an interpretable control variable, to lift the black-box barrier to clinical deployment.

Projects

Sign2Speak | *Keras, MediaPipe, TFLite / LiteRT, Gemma 4 E2B, Android* May 2026 – Present

- Fully on-device American Sign Language translator (50 signs, extensible to 250), built solo for privacy and offline use with no server round-trip.
- Trained a custom Conv1D + Transformer sign classifier (Keras) on the Kaggle ISLR dataset (∼94k sequences, 543 landmarks/frame) and exported it to TFLite via LiteRT for on-device inference.
- Engineered the end-to-end edge pipeline: CameraX → MediaPipe hand/face landmarks → TFLite classifier → Gemma 4 E2B (LiteRT-LM, on-device) turning ASL gloss into fluent English → Android TTS (speech). Built for the Google DeepMind Gemma 4 Good hackathon.

Clinical ML, Endocrine Therapy Toxicity Prediction | *Python, pandas, scikit-learn* September 2025 – Present

- Predicting 1-year toxicities of breast-cancer endocrine therapy. Team project on a de-identified oncology cohort (2,115 patients, 199 variables).
- Established the missing-data mechanism (MAR), benchmarked missForest against MICE, and restricted the cohort to patients with an observed target to avoid imputing the label.

Loodo | *Flutter, Supabase, LLM agent, QLoRA* September 2025 – May 2026

- Social productivity app (a "Strava for work") with a prompt-engineered AI coaching agent (LLM API) able to perform real in-app actions. Prototyped an on-device coach by fine-tuning Mistral 7B with QLoRA (Unsloth).

Education

ESIEA Paris, France
Engineering degree (Master level), major in Artificial Intelligence & Data Science *2022 – 2027 (Expected: 09/2027)*

Tamkang University New Taipei, Taiwan
Computer Science & Information Engineering, University exchange; coursework in Math for ML *2024 – 2025*

Technical Skills

Languages: Python, SQL, Dart
ML / DL: PyTorch, TensorFlow / Keras, Hugging Face Transformers, scikit-learn, Gensim (Word2Vec / SGNS), Unsloth / QLoRA; RecSys - Prod2Vec, Meta-Prod2Vec, StarSpace, negative sampling, UMAP, spectral clustering
On-device / Edge ML: TFLite / LiteRT, LiteRT-LM, MediaPipe, Android (CameraX), Gemma
Data / Tools: Pandas, NumPy, Git, Supabase, Flutter, Google ADK, missForest / MICE
Spoken: English (professional), French (native), Chinese (beginner)

Leadership & Community

GDG on Campus ESIEA, Co-Lead (July 2025 – Present). Launched the chapter, gave a talk on AI agent architecture and led a hands-on workshop.